The Frequency-Tunable Superconducting-Qubit Flux Sensor: Measurement Model with Complete $(\boldsymbol{c}, \boldsymbol{s})$ Delineation

Based on Danilin, Nugent & Weides (arXiv:2211.08344v4, 2024)
augmented with standard superconducting-qubit sensor literature

April 2026

Abstract

We abstract the measurement model of Danilin, Nugent & Weides (2024) into the joint-design language of AdaptiveDesign: an unknown state $\boldsymbol{x}$ (the external magnetic flux threading a SQUID loop), a tunable sensor control $\boldsymbol{s}$ (delay time, repetitions, flux set-point, drive frequency, probe-state architecture), and a fixed device geometry $\boldsymbol{c}$ (transmon and flux-bias-line geometry, Josephson-junction parameters, readout-chain parameters, environmental noise amplitudes). The Ramsey-fringe likelihood $P_{|1\rangle}(\Phi, \tau; \boldsymbol{c})$ is a concrete, physically grounded forward model. The paper performs a reduced one-dimensional bi-level optimization over $f_q^{\max}$ with a prescribed Kitaev phase-estimation inner rule; a full joint-$(\boldsymbol{c}, \boldsymbol{s})$ framework would promote the full circuit geometry to the outer loop while preserving the same inner structure. We tabulate all parameters unambiguously to enable downstream use.

1 Physical setup and target

A frequency-tunable x-mon transmon qubit (or an N-qubit register prepared in an entangled state) is used as a magnetic-flux sensor. The unknown target is the external flux threading the SQUID loop

$$\boldsymbol{x} \;=\; \Phi_{\text{ext}} \;\in\; [0,\, 0.5]\,\Phi_0, \tag{1}$$

restricted to the first half-period of the flux-tunable qubit spectrum. The flux Φ_{ext} enters the qubit Hamiltonian through the split-junction SQUID loop, modulating the Josephson energy and hence the qubit $|0\rangle \to |1\rangle$ transition frequency

$$f_q(f_q^{\max}, \Phi) \;=\; \left(f_q^{\max} + \frac{E_C}{h}\right)\sqrt{\left|\cos(\pi\Phi/\Phi_0)\right|} \;-\; \frac{E_C}{h}, \tag{2}$$

where $f_q^{\max}$ is the maximal transition frequency (set by E_J), E_C is the charging energy, and $\Phi_0 = h/2e$ is the superconducting flux quantum.

2 Ramsey-fringe forward model (the likelihood)

The sensing primitive is a Ramsey pulse sequence:

1. prepare $|0\rangle$;
2. apply $\pi/2$ pulse at drive frequency ω_d to produce $(|0\rangle + |1\rangle)/\sqrt{2}$;
3. free evolution for delay time τ accumulates a flux-dependent phase $\phi(\Phi) = \Delta\omega(\Phi)\cdot\tau$, where $\Delta\omega(\Phi) = \omega_q(\Phi) - \omega_d$;
4. apply a second $\pi/2$ pulse;
5. projectively readout qubit population in $|1\rangle$.

Including decoherence and thermal occupation, the expected $|1\rangle$-population is

$$\boxed{P_{|1\rangle}(\Phi,\tau;\,\boldsymbol{c}) \;=\; \frac{1}{2} \;+\; \frac{1}{2}\,\underbrace{\frac{e^{hf_q/k_BT}-1}{e^{hf_q/k_BT}+1}}_{\text{thermal factor}}\;\underbrace{e^{-A\tau-B^2\tau^2}}_{\text{decoherence envelope}}\;\cos\!\big(\Delta\omega(\Phi)\,\tau\big)} \tag{3}$$

with decoherence coefficients

$$A \;=\; \tfrac{1}{2}\big(\Gamma_1^{\text{cav}}+\Gamma_1^{\text{ind}}+\Gamma_1^{\text{cap}}\big) \;+\; \pi^2 A_\Phi^2\,\big|\partial^2\omega_q/\partial\Phi^2\big|, \tag{4}$$

$$B \;=\; \sqrt{A_\Phi^2\,\big|\partial\omega_q/\partial\Phi\big|^2 \;+\; A_{I_c}^2\,\big|\partial\omega_q/\partial I_c\big|^2}. \tag{5}$$

All five relaxation/dephasing rates $\Gamma_1^{\text{cav}},\Gamma_1^{\text{ind}},\Gamma_1^{\text{cap}},\Gamma_\phi^{\text{flux}},\Gamma_\phi^{\text{curr}}$ depend on the device design parameters $\boldsymbol{c}$ (circuit geometry, junction parameters) and on the operating flux Φ through the expressions given in the paper (Eqs. 2–10).

Entangled N-qubit GHZ probe. Preparing a GHZ state $(|00\cdots0\rangle+|11\cdots1\rangle)/\sqrt{2}$ and projecting to the first qubit at the end yields (Eq. 22 of the source)

$$P^{(N)}_{|10\cdots0\rangle}(\Phi,\tau) \;=\; \tfrac{1}{2}+\tfrac{1}{2}\big(\text{thermal}_N\big)\,e^{-N(A\tau+B^2\tau^2)}\,\cos(N\Delta\omega\,\tau), \tag{6}$$

yielding an N-fold speed-up in phase accumulation at the cost of an N-fold acceleration of the decoherence envelope (under uncorrelated noise; correlated-noise scaling can be faster).

Single-shot statistics. Each Ramsey run produces a binary outcome $y\in\{0,1\}$ distributed Bernoulli with mean $P_{|1\rangle}$. With K repetitions at fixed (Φ,τ) the empirical mean $\hat{P}=K^{-1}\sum_k y_k$ has sampling-noise width $\sigma_P=1/(2\sqrt{K})$ from the central limit theorem; standard dispersive single-shot readout introduces an additional state-assignment error ε (the source uses $\varepsilon=0.01\%$ together with Gaussian discrimination widths $\sigma_0=\sigma_1=1.0$).

3 Fixed parameters $\boldsymbol{c}$: device, hardware, environment

These are set at fabrication or by the experimental environment and do *not* change during a sensing run. They are the "geometry" that a joint-design framework would optimize over. Values in parentheses are those used in the source paper.

3.1 Transmon circuit geometry (Appendix A, Fig. A1 of the source)

Symbol	Value	Meaning
w	$24\,\mu$m	x-mon arm width
L	$130\,\mu$m	x-mon arm length
s	$24\,\mu$m	x-mon electrode to ground-plane gap
a, b, h	90, 45, 6 μm	resonator coupler geometry
x_a	$24\,\mu$m	flux-bias-line arm length
—	5, 2, 2 μm	flux-bias-line widths (parts A, B, C)
SQUID loop area	$125+241.5\,\mu\text{m}^2$	two-rectangle flux-collection loop

3.2 Derived / computed circuit parameters

Symbol	Value	Derivation
C_{qg}	76 fF	x-mon to ground plane (Maxwell FEM in COMSOL)
C_{qr}	2 fF	x-mon to resonator coupler
C_c	0.2 fF	x-mon to xy-control line
β	0.03	voltage divider (xy coupling strength)
E_C/h	0.254 GHz	charging energy, from $C_{qg} + C_{qr} + C_c$
M	2.08 pH	mutual inductance flux-bias $\leftrightarrow$ SQUID loop (Biot–Savart)
M'	0.22 pH	parasitic mutual to total x-mon circuit

3.3 Josephson and junction parameters

Symbol	Value	Meaning
$f_q^{\max}$	**design variable,** 2–20 **GHz**	maximal 0–1 transition frequency (set by E_J)
E_J	implicit via $f_q^{\max}$	Josephson energy of the junctions
$L_J(f_q^{\max}, \Phi)$	Eq. 5 of source	Josephson inductance

3.4 Readout chain

Symbol	Value	Meaning
κ	0.5 MHz	resonator total decay rate
$\Delta = \omega_q - \omega_r$	$2\pi \times 2$ GHz	qubit–resonator detuning (held constant)
Z_0	50 Ω	resonator and control-line characteristic impedance
$g_{01}(f_q^{\max}, \Phi)$	Eq. 3 of source	qubit–resonator coupling

3.5 Environmental noise model

Symbol	Value	Role / spectrum
A_Φ	$10^{-6}\,\Phi_0$	flux-noise $1/f$ PSD amplitude: $S_\Phi(\omega) = 2\pi A_\Phi^2/\omega$
A_{I_c}	$10^{-6}\,I_c$	critical-current $1/f$ PSD amplitude
T	20, 40, 60 mK	sensor temperature (dilution-refrigerator mixing chamber)

3.6 Readout / shot-noise model

Symbol	Value	Meaning
σ_0, σ_1	1.0 each	Gaussian widths for $\|0\rangle/\|1\rangle$ readout histograms
ε	0.01%	state-assignment error probability
$\tau_{\min}$	(hardware-set)	minimum delay between control pulses; limits dynamic range via $\Phi_{\max} = \pi/(\tau_{\min}\|a$
T_r	(hardware-set)	one state-prep + measurement + reset cycle time

3.7 Items not explicit in the paper but needed for a full model

The following are standard in the broader superconducting-qubit literature and would need to be carried in any complete forward model:

- TLS (two-level-system) bath coupling in the dielectric, setting the residual T_1 floor captured in the paper's Γ_1^{cap}.
- Quasi-particle tunneling rate (paper dismisses as Hz-scale; comprehensive models still carry it).

- Asymmetric single-shot readout fidelity $F_r = 1-\varepsilon_{01}-\varepsilon_{10}$ (usually state-specific in practice).
- Residual thermal occupation of the readout resonator (Bose factor $\bar{n}_{\mathrm{th}}$), adding a Purcell-enhanced thermal contribution to A.
- $1/f$ charge noise—paper correctly dismisses in the deep-transmon regime $E_J/E_C \gg 50$.

4 Tunable parameters $\boldsymbol{s}$: per-shot and per-protocol controls

These are the quantities the Kitaev PEA (or any adaptive policy) varies from shot to shot, or from step to step.

4.1 Per-step tunables (adaptive variables)

Symbol	Range	Role
τ_k	$[\tau_{\min}, \sim T_2]$	**principal adaptive variable**; Kitaev PEA doubles $\tau_k = 2^{k-1}\tau_{\min}$
n_k	integer ≥ 1	number of Ramsey repetitions at step k; termination when posterior confidence excee
Φ_{ext}	$[0, \Phi_0/2]$	operating-point offset; in the paper held at Φ^* for the whole run, but in principle retu

4.2 Per-protocol tunables (chosen once per sensing run)

Symbol	Range	Role
Φ^*	in $[0, \Phi_0/2]$	optimal sensing flux maximizing $\partial P_{\|1\rangle}/\partial\Phi$ (Eqs. 12, 15, 18). Paper fir
ω_d	$\sim \omega_q(\Phi^*)$	microwave drive frequency for $\pi/2$ pulses; determines $\Delta\omega$
N	1, 2, 3 (demonstrated)	number of probe qubits (entangled or unentangled)
probe state	$\|+\rangle$, $\|\Phi^+\rangle$, GHZ_N	initial coherent superposition for phase accumulation
PEA variant	Kitaev / Fourier	phase-estimation algorithm (paper uses Kitaev)

4.3 Per-pulse tunables (not explicit in the paper but present in practice)

A more general adaptive design would include:

- $\pi/2$ pulse amplitude and duration (gate fidelity, leakage to $|2\rangle$);
- pulse shape (DRAG-corrected Gaussian vs. square) — affects leakage, Z-errors;
- readout pulse amplitude and duration — trades measurement back-action for fidelity;
- active reset vs. passive thermalization between shots, affecting T_r;
- for multi-qubit: CP_{ij} conditional-phase gate parameters (pulse timing, flux-pulse amplitude).

These become relevant $\boldsymbol{s}$ parameters once one optimizes beyond the idealized phase-accumulation picture used in the source.

5 What the paper optimizes, and what a joint-design framework would add

5.1 What the paper explicitly optimizes

The source performs a bi-level optimization in the spirit of path (2a)/(2b) of the relaxation hierarchy (AdaptiveDesign, §10):

- **Outer (design):** scan over $f_q^{\max} \in [2, 20]$ GHz, a *one*-dimensional slice of $\boldsymbol{c}$. For each value, find the optimal sensing flux $\Phi^*(f_q^{\max})$ analytically by maximizing single-shot sensitivity $S(f_q^{\max}, \Phi) = \max_\tau |\partial P_{|1\rangle}/\partial\Phi|$ (Eqs. 16, 18).

- **Inner (policy):** Kitaev PEA prescribes τ_k and n_k from the current belief at run time. The delay-time schedule $\{\tau_k\}$ is the adaptive variable; the termination and majority-voting rules at each step are shot-level adaptive.

This is a clean instance of a prescribed-inner-rule bi-level loop (path (2b), myopic/information-theoretic with a multi-step algorithm template) rather than a full Bellman-optimal policy over the Ramsey likelihood.

5.2 What the paper does *not* vary (but a full joint framework could)

- **Circuit geometry** (w, L, s, coupler geometry, flux-bias-line widths, SQUID-loop shape). These control M, M', C_{qg}, C_{qr}, C_c, β, and hence every Γ rate; they enter the likelihood through the A, B envelope of Eq. (3).
- **Operating temperature** T — varied for comparison at 20, 40, 60 mK but not treated as a design variable (though constrained by the mixing-chamber budget).
- **Qubit–resonator detuning** Δ — fixed at $2\pi \times 2\,\mathrm{GHz}$. Purcell decay $\Gamma_1^{\mathrm{cav}} \propto 1/\Delta^2$, so larger Δ improves coherence at the cost of slower readout.
- **Resonator design** (κ, Q_e, Q_i) — fixed.
- **Multi-qubit placement and mutual couplings** — the number N is varied, but the geometric layout of the N qubits and their couplings is not.
- **Flux-bias-line / SQUID-loop geometry**, which simultaneously controls the inductive Purcell channel Γ_1^{ind} and the sensor's dynamic range $\Phi_{\max}$ (Eq. 20).
- **Probe-state preparation sequence** — hand-designed ($|\Phi^+\rangle$ Bell / GHZ via CP_{10}) rather than optimized.

5.3 A complete joint-$(\boldsymbol{c}, \boldsymbol{s})$ formulation

In the notation of AdaptiveDesign:

$$\boldsymbol{x} = \Phi_{\mathrm{ext}}, \tag{7}$$

$$\boldsymbol{s}_k = \big(\tau_k,\ n_k,\ \Phi^*,\ \omega_d,\ N,\ \text{probe state}\big), \tag{8}$$

$$\boldsymbol{c} = \big(f_q^{\max},\ \text{circuit geometry},\ \text{coupler/resonator design},\ T,\ \Delta\big), \tag{9}$$

with per-shot likelihood

$$p(y_k \mid \boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c}) = P_{|1\rangle}\big(\Phi_{\mathrm{ext}}, \tau_k;\, \boldsymbol{c}\big)^{y_k} \left(1 - P_{|1\rangle}\right)^{1-y_k}, \tag{10}$$

for $P_{|1\rangle}$ given by Eq. (3) (or Eq. (6) in the GHZ case) and with the K-shot empirical mean $\hat{P}$ carrying Gaussian sampling noise $\sigma_P = 1/(2\sqrt{K})$ modulated by the ε readout-error model.

A full joint DP would

- optimize the circuit geometry $\boldsymbol{c}$ at the outer level, with envelope-theorem gradients propagating through the FEM capacitance extraction (COMSOL, or a differentiable surrogate) and the Biot–Savart inductance integral;
- optimize the per-step controls (τ_k, n_k, $\Phi_{\mathrm{ext},k}$) at the inner level; and
- optionally treat the probe-state structure (N, entangler choice) as a discrete design layer.

The paper solves a reduced problem: one-dimensional outer scan over $f_q^{\max}$; prescribed Kitaev PEA on τ; fixed geometry; fixed entangler structure within each comparison.

6 Position in the relaxation hierarchy

Within the tractability hierarchy of AdaptiveDesign §10.4, the Danilin–Nugent–Weides pipeline sits between rows (2a) and (2b):

- **Inner rule.** Kitaev PEA — a binary-search-like reduction of the flux interval with doubling τ. Non-myopic (the algorithm has a fixed multi-step template) but *not* Bellman-optimal: the PEA schedule is fixed in advance once $\tau_{\min}$ and N_{step} are set; only the majority-voting termination is adaptive at the shot level.
- **Outer loop.** Grid scan over $f_q^{\max}$; Φ^* found analytically via Eq. 16 of the source.
- **Gradient structure.** Not explicitly formulated in the source; implicit via max-over-τ and max-over-Φ at each $f_q^{\max}$. A joint framework would propagate $\partial S/\partial$(circuit geometry) through the full FEM + Biot–Savart chain, treating $\boldsymbol{c}$ as a ~ 10-dimensional geometric vector.

This makes the paper a clean *one-dimensional worked example* of the bi-level path. The Ramsey likelihood (3) and the GHZ likelihood (6) are tractable, physically grounded forward models. The system slots directly into the joint-$(\boldsymbol{c}, \boldsymbol{s})$ framework as an inner-rule choice (path (2b)) with the x-mon circuit geometry promoted from "fixed at fabrication" to "design variable in $\mathbb{R}^{d_c}$."